CIPROFLOXACIN

Tendon rupture risk

Separate from Ca/Fe/antacids

CHEMICAL NAME 1-cyclopropyl-6-fluoro-4-oxo-7-(piperazin-1-yl)-1,4-dihydroquinoline-3-carboxylic acid (as hydrochloride)

CLASS Fluoroquinolone antibiotic (second generation)

TRADE NAMES Ciprobay · Cipro · Ciproxin · Ciprodar

THERAPEUTIC USES / INDICATIONS

- Complicated UTI and pyelonephritis; prostatitis.
- Gram-negative infections incl. *Pseudomonas* (only oral antipseudomonal).
- Typhoid fever, infectious diarrhoea (travellers', shigellosis).
- Bone and joint infections; anthrax; gonorrhoea (if susceptible).
- Eye/ear drops for bacterial conjunctivitis, otitis externa.

ADMINISTRATION

- **PO:** with water; 2 h before or 6 h after antacids, calcium, iron, zinc, milk/dairy alone.
- **IV:** infusion over 60 min into large vein.
- Encourage fluids (prevent crystalluria).

DOSAGES

- **Adult PO:** 250–750 mg every 12 h.
- **Uncomplicated UTI:** 250 mg every 12 h for 3 days.
- **IV:** 400 mg every 8–12 h.
- **Children (specific indications only, e.g. CF, anthrax):** 10–20 mg/kg every 12 h.
- CrCl <30: halve dose.

PHARMACOKINETICS

- **Peak:** PO 1–2 h. Bioavailability ~70%.
- Partly metabolised in liver; inhibits CYP1A2.
- Excreted in urine. **t½:** 4 h.

PHARMACODYNAMICS

- Inhibits bacterial **DNA gyrase (topoisomerase II) and topoisomerase IV** → bactericidal.
- Strong Gram-negative cover incl. *Pseudomonas*; weak against streptococci and anaerobes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea, abdominal pain.
- Headache, dizziness, insomnia.
- Photosensitivity; rash.

ADVERSE EFFECTS (SERIOUS)

- Tendinitis and tendon rupture (Achilles; risk ↑ age >60, steroids, transplant).
- QT prolongation; aortic aneurysm/dissection.
- Peripheral neuropathy; CNS effects (seizures, psychosis, confusion); myasthenia exacerbation.
- Dysglycaemia; *C. difficile* colitis.

CONTRAINDICATIONS & CAUTIONS

- Fluoroquinolone hypersensitivity; history of FQ tendon disorder.
- Use with **tizanidine**.
- **Caution:** children and pregnancy (cartilage), epilepsy, myasthenia gravis, long QT, aneurysm risk, elderly on steroids.
- Reserve for infections where alternatives are unsuitable (safety warnings).

KEY INTERACTIONS

- **Antacids, Ca, Fe, Zn, sucralfate, dairy:** ↓ absorption.
- **Theophylline, tizanidine, caffeine:** ↑ levels (CYP1A2) → toxicity.
- **Warfarin:** ↑ INR.
- **Steroids:** ↑ tendon rupture; **NSAIDs:** ↑ seizure risk.

NURSING MANAGEMENT

- Separate from minerals and antacids.
- Assess tendon pain, especially in elderly and those on steroids.
- Monitor blood glucose in diabetics.
- Monitor mental status, ECG risk factors.
- Encourage fluids; monitor urine output.

PATIENT EDUCATION

- Stop and report any tendon pain or swelling; rest the limb.
- Avoid strong sun; use sunscreen.
- Do not take with milk, antacids, iron or calcium.
- Report tingling, confusion, palpitations.

KEY POINTS

- Fluoroquinolone; DNA gyrase inhibitor.
- Tendon rupture, QT, CNS effects.
- Only oral drug covering Pseudomonas.
- Chelates with Ca, Mg, Fe, Zn.